

Human DGAT1 (UniProt O75907) - Quick Reference

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Basic Information

Property	Value
UniProt ID	O75907
Gene	DGAT1
Protein Name	Diacylglycerol O-acyltransferase 1
Length	488 amino acids
MW	55,278 Da
EC Number	2.3.1.20

PDB Structures

PDB ID	Resolution	Content
6VP0	3.10 Å	DGAT1 + oleoyl-CoA
6VYI	3.00 Å	Apo form
6VZ1	3.20 Å	DGAT1 + acyl-CoA
8ESM	3.20 Å	DGAT1 + T863 inhibitor
8ETM	3.20 Å	DGAT1 + ligand
AlphaFold	-	AF-Q75907F

Topology (9 Transmembrane Helices)

[1-83] Cytoplasmic (N-terminal) [84-118] TM1 ⌋
[119-130] | Lumenal [131-156] ⌋ TM2 [157-161] |
Cytoplasmic [162-184] ⌋ TM3 [185-191] | [192-223] ⌋

TM4 [224-273] | Lumenal [274-308] ┤ TM5 [309-315] |
Cytoplasmic [316-353] ┤ TM6 [354-399] | Lumenal
[400-420] ┤ TM7 [421-428] | Cytoplasmic [429-447] ┤
TM8 [448-449] | Catalytic loop [450-481] ┤ TM9
[482-488] C-terminal (Cytoplasmic)

Catalytic Residues

Position	Residue	Role
ACT_SITE 415	His	Catalytic (essential for activity)
SITE 416	Adjacent	Catalytic dyad

Key His residue (Literature: His449) = Invariant catalytic residue

MBOAT Family

DGAT1 belongs to **Membrane-Bound O-Acyltransferase (MBOAT)** family

Feature	Description
MBOAT fold	Hollow chamber in membrane
Catalytic chamber	Conserved His/Asn residues
Acyl-CoA tunnel	Cytosol-facing, ~15-20 Å
DAG cavity	ER membrane/lumen-facing

Key Binding Sites

Acyl-CoA Binding Site

- Location: Cytosolic-facing tunnel
- Key residues: Catalytic His415/449, aromatic residues (Phe, Tyr)
- Inhibitors: T863, A-922500, DGAT1IN1

DAG Binding Site

- Location: ER membrane/lumen-facing cavity
 - Substrate access for diacylglycerol
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Inhibitor Binding Modes

Inhibitor	PDB	Binding Mode
T863	8ESM	Tunnel entrance blocking
DGAT1IN1	8ESL	Deeper penetration, amide H-bond to catalytic His
Oleoyl-CoA	6VZ1	Natural substrate, CoA moiety binds cytosolic side

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